# KARTHIK M

Lead Software Architect | Backend, Distributed Systems, AI in Production | Java, Go, Python

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## PROFESSIONAL SUMMARY

Nine years in backend engineering and two and a half as an architect, owning the platform behind a regulated business and the engineers who build it.

## KEY ACHIEVEMENTS

- **Platform owned end to end.** An estate of 54 Go and Java microservices on Temporal, inherited with overlapping responsibilities and most original authors gone, moved onto a coherent target architecture incrementally with no business disruption.
- **AI in production, under governance.** Two systems live. One runs member administration from the inbox and cut onboarding time 40%. The other adjudicates claims, retrieving line items through an embedding-based vector search layer and deciding admissibility and the payable amount. Every automated decision held behind maker and checker approval.
- **Direction set beyond the reporting line.** A gateway at 50,000+ calls a minute and 99.99% availability, a DNS security platform at 7B+ queries a day, and a broken authorisation layer rebuilt from 90+ production incidents a day to zero. Architecture decisions and design review carried well beyond any direct line, with trade-offs explained to product, operations and external partners.

## SKILLS

- **Languages:** Java, Spring Boot, Go, Python, FastAPI, Node.js, C#, REST, gRPC, GraphQL
- **Cloud and Platform:** AWS, Lambda, S3, ECS, Kubernetes, Docker, Terraform, CI/CD, Grafana
- **Data and Messaging:** PostgreSQL, MySQL, MongoDB, Redis, Kafka, Temporal, Elasticsearch
- **AI and Security:** LangChain, LangGraph, MCP, SAML 2.0, OAuth 2.0, OIDC, SCIM 2.0, RBAC

## PROFESSIONAL EXPERIENCE

### Lead Software Architect | Onsurity

Jul 2022 - Present | HealthTech & InsurTech, India

- Owned EDITH end to end, the platform behind the business: 54 Go and Java microservices on Temporal spanning API gateway, authentication, identity and RBAC, underwriting, policy issuance, claims adjudication with cashless and reimbursement settlement, insurer and TPA integration, billing and wallet, across group health, life, personal accident and commercial lines, with 7.5M registered users on it and 200,000 gig workers onboarded and offboarded every day.
- Built the onboarding AI that runs member administration from the inbox: it reads inbound customer email, decides whether the request is an enrolment, a deletion or an endorsement, maps it to the right API contract and calls the platform. Where validation fails it drafts a reply naming the fields to correct. Onboarding time fell 40%.
- Built the claims AI that adjudicates end to end: uploaded documents are classified, line items retrieved through an embedding-based vector search layer, and each compared against the policy wording to decide admissibility and compute the payable amount. Fraud triggers are raised from ailment and treatment combinations and prior hospital history. Both run under maker-checker governance.
- Re-architected the company-wide platform. The estate had grown with overlapping responsibilities and most original authors gone. Defined a coherent target architecture and moved to it incrementally with no business disruption, running old and new paths side by side until outputs matched. Replaced a third-party telephony provider with Astrix, an in-house calling platform built in Go.
- Took the company's B2B2C insurance line from zero to production: the multi-tenant retail platform carrying 2M+ lives under active cover with per-tenant isolation, RBAC and doctor consultation, and the B2B distribution surface for insurers, TPAs and broker networks. Built the money layer with it: payment gateway, billing and invoicing, and settlement and reconciliation across partners, with actuarial pricing models converted from spreadsheets into callable FastAPI services and materialised as 80M pre-computed rows, cutting quote generation effort by 60%.
- Led the engineering workstream for SOC 2 and ISO 27001 certification, both achieved: access control and RBAC, per-tenant data isolation, audit trails on consequential actions, and data protection aligned to GDPR.

- Migrated broken legacy identity and RBAC onto a new implementation, taking 90+ production incidents a day to zero, and absorbed two teams totalling 14 engineers with the platforms in scope: API gateway, authentication, notifications and billing.
- Built a secure multi-tenant Broker Management System with per-tenant data isolation, cutting API response times 40% through query optimisation, caching and service restructuring.
- Built ClaimX, end-to-end claims processing with third-party administrator integrations over message queues, designed for idempotent processing so a redelivered event never settles a claim twice, alongside the Insurer Enrollment Service for certificate-of-insurance generation and state consistency across insurer partners, Loansure, an emergency lending microservice with automatic settlement recovery, and an OCR document-extraction service cutting processing latency 35%.
- Led a 23-engineer group of 12 backend, 4 AI/ML, 2 frontend and 5 QA engineers, owning hiring, onboarding, performance reviews and promotions, with standards adopted throughout the 45-person engineering function of a 300+ employee company. Owned target architecture and build-versus-buy decisions, and drove organisation-wide adoption of agentic development tooling with guardrails and training.

*Tech: Go, Java, Spring Boot, Python, FastAPI, Node.js, REST, gRPC, GraphQL, OpenAPI, Temporal, Kafka, MongoDB, PostgreSQL, MySQL, Redis, Elasticsearch, Docker, Kubernetes, Terraform, Grafana, AWS, LangChain, LangGraph, ElevenLabs*

**Software Development Engineer II | Halodoc** Apr 2022 - Jun 2022 | Bengaluru, India

- Built microservices for patient consultation and appointment booking handling 10,000+ daily consultations at Indonesia's largest healthtech platform (20M+ users a month), plus a HIPAA-compliant EHR system and a WebRTC telemedicine feature cutting wait times 40%.

*Tech: Java, Spring Boot, Kafka, Redis, MongoDB, WebRTC, AWS*

**Software Engineer | Infoblox** Mar 2020 - Mar 2022 | Bengaluru, India

- Built the identity platform end to end in Go, the layer every customer was identified through: the REST API gateway fronting DNS and DHCP management at 50,000+ calls a minute and 99.99% availability, secured with OAuth 2.0 and OpenID Connect, with SCIM 2.0 provisioning endpoints serving Okta, Azure AD and Ping, and versioned contracts consumed two years with no breaking change.
- Developed a distributed DNS security platform in Go providing DDoS and DNS-layer threat protection, and took it to production at 7B+ queries a day, where distributed caching and load-balancing cut query response times by 60%.

*Tech: Go, Java, AWS, Azure AD, Okta, OAuth 2.0, OIDC, SCIM 2.0, Redis, PostgreSQL, Docker, Kubernetes, Grafana, DNS*

**Software Engineer II | Goldman Sachs (via Saggezza)** Nov 2019 - Mar 2020 | Bengaluru, India

- Implemented and optimised an Elasticsearch cluster for financial data indexing, supporting trading and analytics platforms.

*Tech: Elasticsearch, Java, Spring Boot*

**Software Developer | Ultria** Jul 2017 - Nov 2019 | Bengaluru, India

- Built the SAML 2.0 multi-tenant single sign-on module for a contract lifecycle product used by 50+ Fortune 500 clients, federating with each client's own IdP, with tenant isolation and RBAC throughout.
- Owned search across millions of contracts: built on the Lucene API with custom analysers and relevance tuning, then led the migration to Elasticsearch, improving search performance by 75%.
- Built the workflow engine behind contract approvals end to end, and WordConnect in C#, a Microsoft Word integration syncing contract actions back into the platform.
- Led a team of 8 from the second year, owning work allocation, design and code review, and technical direction.

*Tech: Java, C#, Spring Boot, Lucene, Elasticsearch, SAML 2.0, SSO, Grafana, REST APIs, AWS*

## EDUCATION

**Bachelor of Engineering in Computer Science** | Visvesvaraya Technological University (VTU) | Aug 2013 - Apr 2017